

Tokenized Asset Risk Metrics (TARM): A Framework for Institutional Risk Assessment in Tokenized Real-World Assets

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Abstract

The rapid growth of tokenized real-world assets has created new opportunities for financial innovation while simultaneously introducing complex structural risks for institutional participants. Existing risk assessment approaches in traditional finance and blockchain research frequently examine individual risk factors in isolation, limiting their usefulness for consistent institutional comparison. This paper introduces Tokenized Asset Risk Metrics (TARM), a transparent, semi-quantitative framework designed to evaluate the institutional suitability of investable tokenized real-world assets at a given point in time. TARM brings together financial, legal and regulatory, operational and custody, and on-chain governance considerations within a unified assessment structure. The framework is intended as a pre-investment screening tool rather than a model for predicting prices, returns, or market performance. Its purpose is to identify structural weaknesses that may affect an asset's ability to satisfy institutional requirements under legal, regulatory, operational, and governance stress. The framework provides a structured basis for comparative assessment across tokenized asset classes and is demonstrated through applications to tokenized gold, tokenized bonds, tokenized real estate, and tokenized equities. The study highlights the importance of legal enforceability, ownership structures, custody arrangements, and governance controls in assessing institutional suitability. By emphasizing transparency, interpretability, and reproducibility, TARM provides a practical foundation for institutional risk screening, comparative analysis, advisory applications, and future regulatory research in tokenized financial markets.

Keywords: Tokenization; Real-World Assets; Institutional Risk; Blockchain Finance; Digital Assets; Tokenized Assets; Risk Assessment

For information regarding the full paper, detailed methodology, assessment instrument, or research materials, please contact:

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Public-disclosure boundary: This public SSRN abstract intentionally provides a high-level description of the research. The detailed scoring instrument, assessment questions, aggregation specification, and implementation details are not reproduced here.